THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): The rate of breathing in aquatic organisms is much faster than in terrestrial organisms. Reason (R): The amount of dissolved oxygen in water is fairly low compared to air. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): Reflex actions are not controlled by the brain directly. Reason (R): The reflex arc connects the receptor to the effector through the spinal cord. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): Asexual reproduction produces offspring that are not exactly identical to the parent. Reason (R): Copying of DNA is accompanied by small errors. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Sex of a child is determined by the father. Reason (R): Sperms carry either X or Y chromosome. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. In aerobic respiration, one molecule of glucose gives how many molecules of carbon dioxide? 1

(a) 2 **(b)** 12 **(c)** 6 **(d)** 1

6. A hormone X is secreted by an organ below the stomach, regulates sugar. X's deficiency causes: 1

(a) goitre **(b)** dwarfism **(c)** diabetes **(d)** gigantism

7. A plant breeder crosses two varieties and also grows a plant by stem cutting. The offspring from the cross show 1

(a) identical traits in the cross
(b) variation in the cutting only
(c) no variation in either
(d) variation, and those from the cutting are identical to the parent

8. A pea plant RrYy is crossed with rryy. Out of 200 offspring, the number of round green plants is expected to be 1

(a) 25 **(b)** 50 **(c)** 150 **(d)** 100

9. A closed jar contains a green plant in sunlight together with a mouse. The mouse survives longer than a mouse alone in a jar because the plant: 1

(a) uses carbon dioxide and releases oxygen **(b)** absorbs oxygen only
(c) releases carbon dioxide **(d)** gives out nitrogen

10. A boy has a rapid pulse, sweating and wide pupils when frightened. This is due to: 1

(a) adrenaline **(b)** thyroxine **(c)** oestrogen **(d)** insulin

11. An organism with a very complex body that can still regenerate would need 1

(a) no cells at all
(b) undifferentiated cells to produce missing parts
(c) only gametes
(d) no DNA

12. A pea plant that is tall and round (TtRr) is crossed with dwarf wrinkled (ttrr). The offspring ratio is 1

(a) 3 : 1 **(b)** 1 : 1 : 1 : 1 **(c)** 9 : 3 : 3 : 1 **(d)** 1 : 1

13. **Assertion (A):** Guard cells control the opening and closing of stomata. 1
Reason (R): Guard cells swell when water flows into them, causing the stomatal pore to open.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** A reflex action is a quick, automatic response. 1
Reason (R): In a reflex action the signal passes through the spinal cord and not through the brain.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** Plants raised by vegetative propagation show characters identical to the parent plant. 1
Reason (R): Vegetative propagation involves only one parent and no fusion of gametes.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Calculate the volume of blood pumped in 1 minute if the heart beats 72 times per minute and pumps 70 mL each time. 2

17. A patient cannot pull his hand away from a hot object but can feel the heat. Which part of the reflex arc is damaged? 2

18. A tiny part of a Hydra body is used to grow a new Hydra in a lab. Is this regeneration or budding? Justify. 2

19. In a cross of two heterozygous tall plants (Tt x Tt), 240 plants were produced. Find the number of tall plants that are homozygous. 2

20. Why is the pH of the small intestine alkaline and why is that necessary? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In an experiment, the destarched leaf was covered with black paper partly. After sunlight and iodine test the uncovered part turned blue-black. Interpret fully. 3

22. A seedling is kept in a box with a window on one side; after a few days it bends. When the tip was cut, no bending was observed. Explain. 3

23. After 2 hours, the number of Amoebae from one cell that divides every 15 minutes is calculated. Find it and state the type of reproduction. 3

24. A pea plant has yellow round seeds. How would you find its genotype? Describe the cross and the possible results. 3

OR

Explain why the human heart muscle needs its own blood supply (coronary arteries). 3

25. A frog's leg muscle twitches when a nerve is stimulated. Explain which nerve and the reflex path. 3

OR

Why does cutting a plant stem and keeping it in moist soil give a new plant while cutting a human finger does not give a new human? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Meera had a lunch of chapatti, dal and a glass of milk. The food first entered her mouth, where it was mixed with saliva, and then passed through the oesophagus into the stomach and later into the small intestine. 5

(i) Name the enzyme in saliva that acts on chapatti.
(ii) Name the acid secreted in the stomach.
(iii) Where does the complete digestion of food take place?
(iv) State the role of bile juice in the digestion of fat in milk and explain why the medium in the small intestine must be alkaline.

27. Explain, with reference to variations, why sexual reproduction is more advantageous to a species than asexual reproduction, giving an example from plants. 5

OR

A pea plant homozygous for tall and round seeds is crossed with a dwarf plant having wrinkled seeds. Show the F1 and the F2 with the number of each phenotype in 1600 F2 plants, and the genotype of the dwarf round plants. 5

END OF QUESTION PAPER — Total: 50 Marks